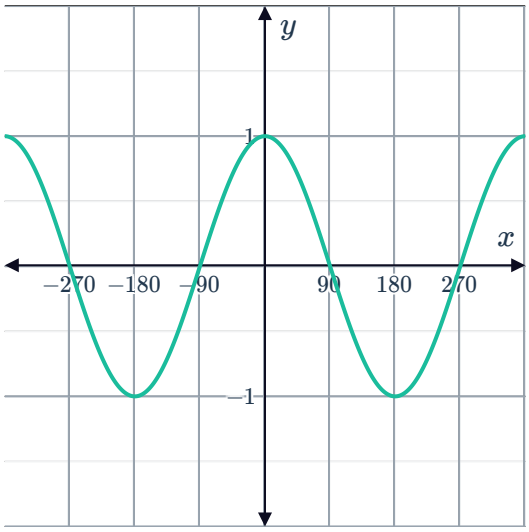
Vertical dilations

1

a

θ	0	60	90	120	180	240	270	300	360
$\cos\theta$	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1

b



c 1

d -1

e $[-4, 4]$

2

a $(-\infty, \infty)$

b $[-1, 1]$

3

a $(-\infty, \infty)$

b $[-2, 2]$

4 8

5

a $y = 2\sin x$

b $y = -\sin x$

c $y = 3\cos x$

d $y = -1.5\cos x$

e $y = 3\sin x$

f $y = 2.5\sin x$

6 270

7

a 8

b 0

c -8

8 5

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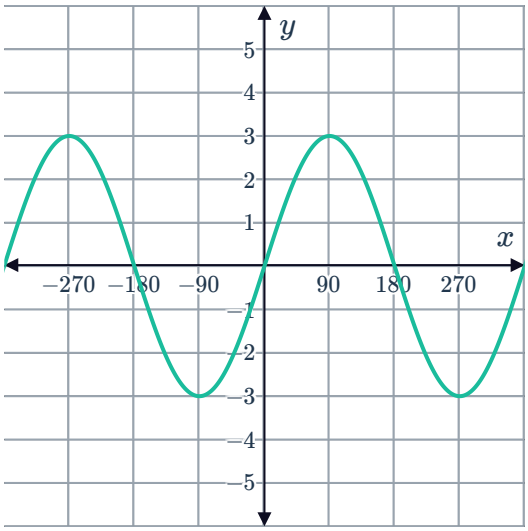
9 $-2\sin x$

10

a

i $A = 3$

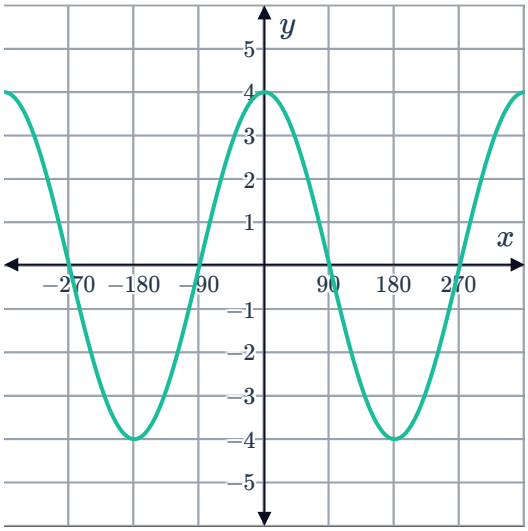
ii



b

i $A = 4$

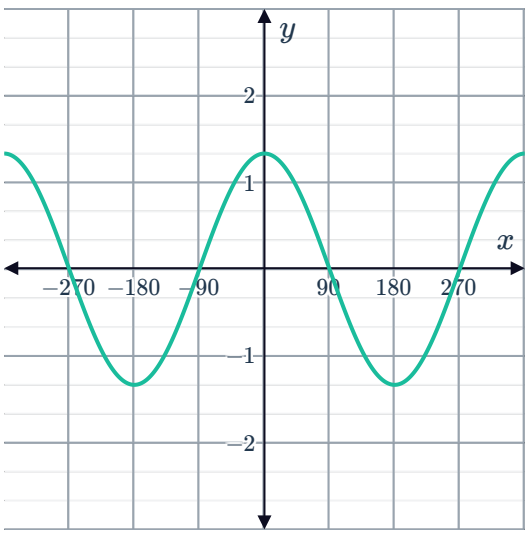
ii



c

i $A = \frac{4}{3}$

ii

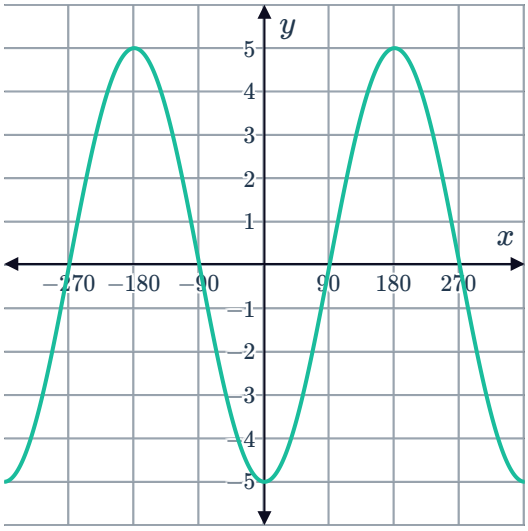


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d

i $A = 5$

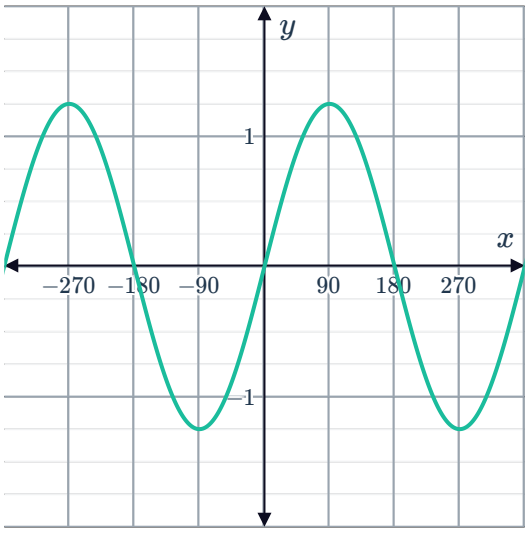
ii



e

i $\frac{5}{4}$

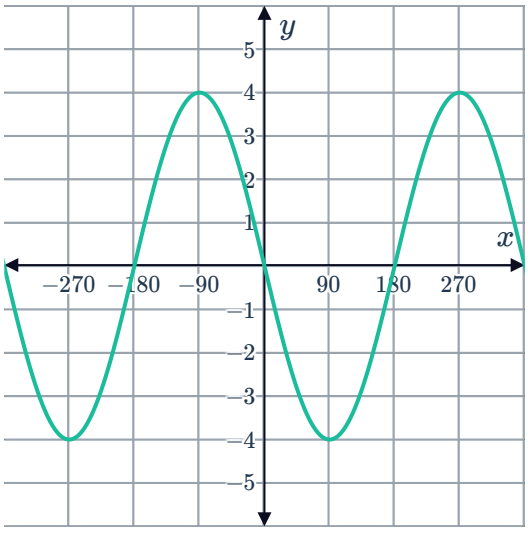
ii



f

i $A = 4$

ii



11

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b -3

c 3

d A vertical dilation by a factor of 3 and reflection about the x-axis.

Vertical translations

12 Translated 2 units downwards

13 Translated 2 units upwards

14 Translated 4 units upwards

15

a $y = \sin x + 4$ **b** $y = \sin x - 2$ **c** $y = \sin x - 4$ **d** $y = \sin x + 1$

16

a $y = \cos x + 3$ **b** $y = \cos x - 4$ **c** $y = \cos x - 3$ **d** $y = \cos x + 2$

17 Reflection about the x-axis, then translation 3 units up.

18 $y = 2$

19

a Translated 4 units upwards **b** 360°

c 5 **d** 3

20

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a

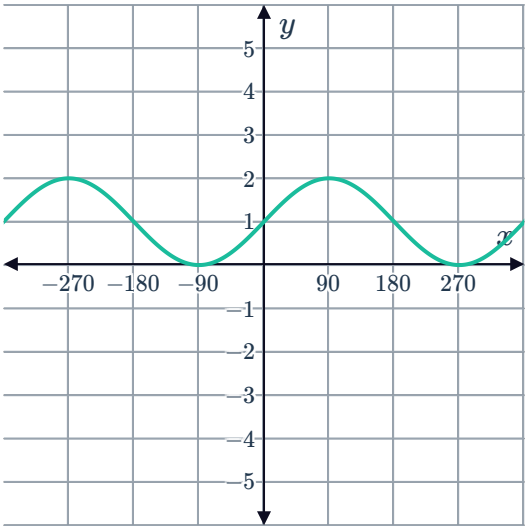
i 360

ii 1

iii 2

iv 0

v



b

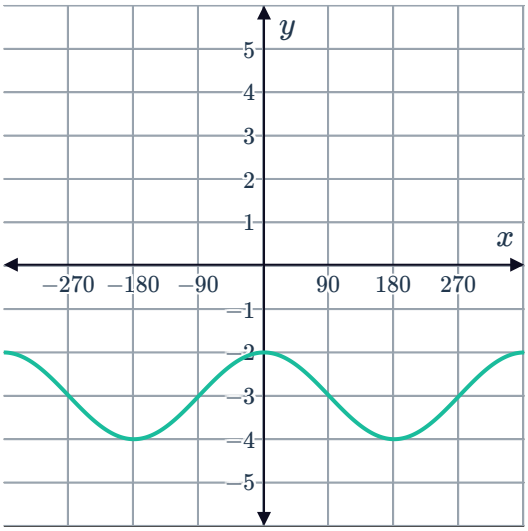
i 360

ii 1

iii -2

iv -4

v



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c

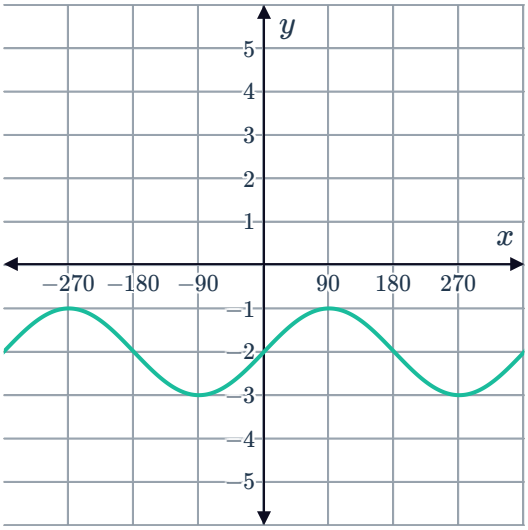
i 360

ii 1

iii -1

iv -3

v



d

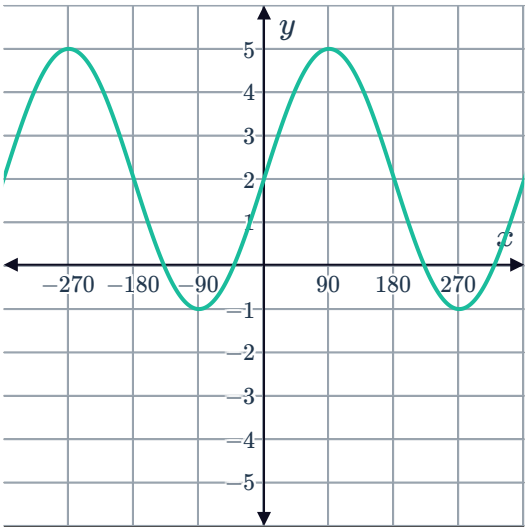
i 360

ii 3

iii 5

iv -1

v



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e

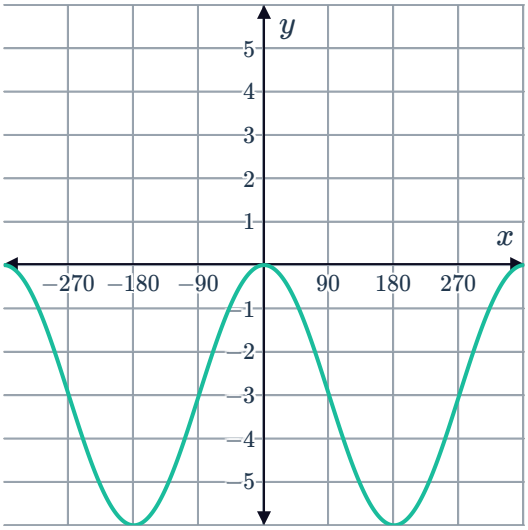
i 360

ii 3

iii 0

iv -6

v



f

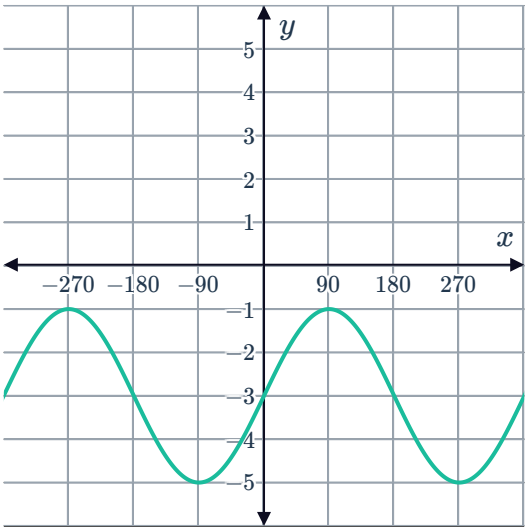
i 360

ii 2

iii -1

iv -5

v



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21

a

i 3

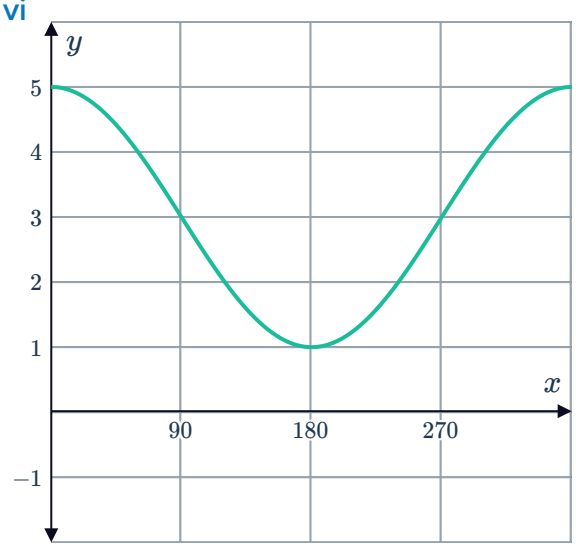
ii 2

iii 5

iv 1

v A vertical dilation by a factor of 2, then translated 3 units upwards.

vi



b

i -3

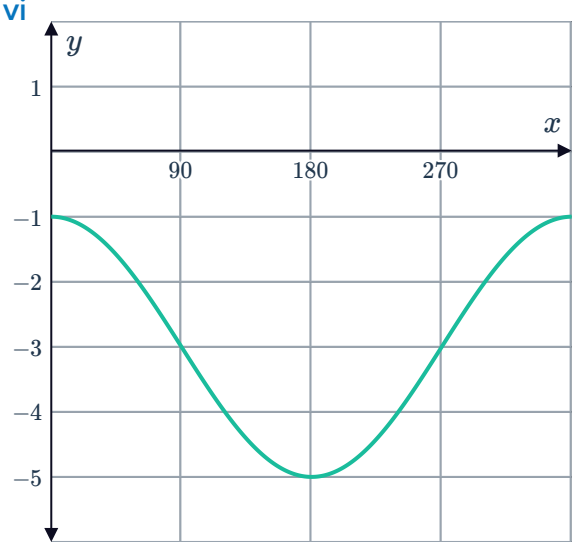
ii 2

iii -1

iv -5

v A vertical dilation by a factor of 2, then translated 3 units downwards.

vi



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c

i 2

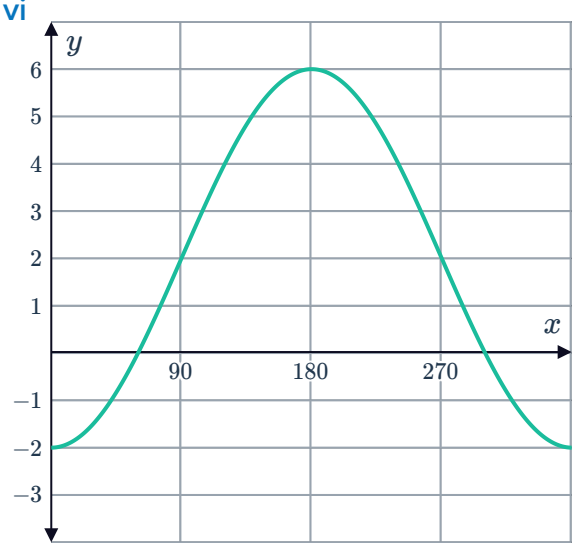
ii 4

iii 6

iv -2

v A vertical dilation by a factor of 4, a reflection about the x-axis and then translated 2 units upwards.

vi



d

i $\frac{7}{3}$

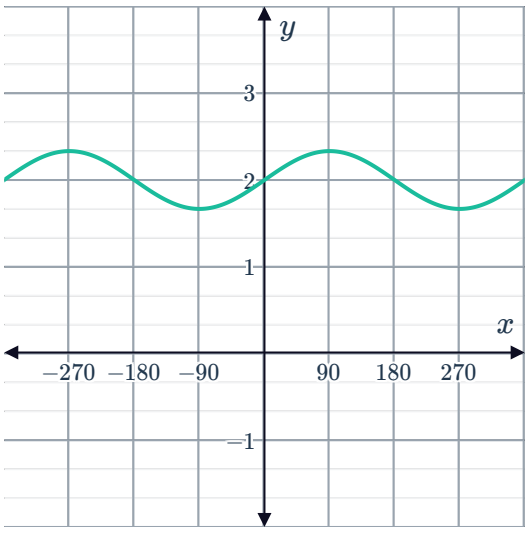
ii $\frac{1}{3}$

iii $\frac{7}{3}$

iv $\frac{5}{3}$

v A vertical dilation by a factor of $\frac{1}{3}$, then translated 2 units upwards.

vi



22

a $y = \cos x + 9$

b 10

23

a i 5 ii -1

b i 7 ii 1

c i 2 ii 1

d i 2 ii $\frac{1}{2}$

24 $2\sin x + 2$

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25 $8\cos x - 2$

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